Experiment 12-Experiment to detect outliers in a given data set.

Description:

Understand the procedure to identify the outliers in a given dataset

PROGRAM:

```
import numpy as np
import matplotlib.pyplot as plt
# Sample dataset
data = np.array([25, 28, 30, 32, 29, 31, 35, 33, 27, 70])
# Calculate Q1 and Q3
Q1 = np.percentile(data, 25)
Q3 = np.percentile(data, 75)
# Calculate IQR
IQR = Q3 - Q1
# Calculate limits
lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR
# Detect outliers
outliers = data[(data < lower_limit) | (data > upper_limit)]
print("Dataset:")
print(data)
print("\nQ1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)
print("\nOutliers:")
print(outliers)
# Box plot
```

```
plt.boxplot(data)
plt.title("Outlier Detection using Box Plot")
plt.ylabel("Values")
plt.grid(False)
plt.show()
```

OUTPUT:

```
Dataset:
[25 28 30 32 29 31 35 33 27 70]

Q1: 28.25
Q3: 32.75
IQR: 4.5

Outliers:
[70]
```

